CORE JAVA

Oops -- Object oriented programming structure

class

method

object

JDK====Java development kit===>JVM+JRE==>configuration/overall process

JVm---java virtual machine----->memory allocation and object creation,garbage collection

JRE---java run time environment-->predefined files or library files

Tools:

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Eclipse---oxygen 3a,IDE,neon

netbeans

ibm

rad

notepad

OOps:

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Object oriented programmimng structure

real time

car(class)--->object(ferrari,audi,bmw)-->(method)set of logic

OOps concept:

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inheritance---->to inherit the parent class property by child class is known as inheritance

polymorphism--->poly--many, morphism--forms-----> same method different forms

abstraction---->we cant create any business logic inside the method

encapsulation-->wrapping of code---private variables access

Hierarchy:

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project

package

class---blueprint

method--task or logic(set of actions)

object--by using obj we can call methods

to create java project:

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file----new---java project---name--finish

right click src(source file)----new---package----name---ok

package--right click---class---name---create

syntax:

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class----public class className{}

method---private void methodName(arguments);

object---className objName=new ClassName();

main method---public static void main(string[] args){}

meth1--sub

meth2--mult

meth3—div

variable:

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All variables must be declared before they can be used

The identifier is the name of the variable

It can initialize the variable by specifying an equal sign and a value

To declare more than one variable of the specified type, use comma

rules to create variable:

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words with upper and lower case

number after the letters

only underscore is allowed while creating variables

No special character are allowed except _

no space is allowed between words

variables starts with number is not applicable

variable declaration:

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datatype varName;

datatype varName=value;

int myvalue;// words can be given

int myvalue=10;

types of variable:

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local variable

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within the block scope

we cant call throughout the program

```
main(){
{
int a;
}
sysout(a);//error
}
```

global varibale

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starting of the line

we can call throughout the program

main{

int a;

{

sysout(a);

}

}

static variable

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method , datatype we can create as static

we can call directly without object

only static variables are allowed to static methods

Example program:

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```
public class IronMan {
              int  a=10;//global variable
              static int c;//static variable


              //static method
private static void spiderMan() {
             //local variable
             int b=10;
             System.out.println("Spider man");
             System.out.println(a);//error(can't call global variable in static method)
             System.out.println(b);
```

```
}
   System.out.println(b);//error---local variable can't be accessed outside the method
public static void main(String[] args) {
        IronMan iM=new IronMan();
        //we can call static method without object
        spiderMan();
}
```

Types of class:

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Final Class-----------> we cant inherit

Static Class---------->we cant create class as static

Abstract Class-------->it may contain abstract methods